

RTD

Ixsander Daems

Simon Tassent

Jarne Lonneux

1 Introduction

In this project, Python code was developed to simulate Resonant Tunnel Diodes (RTDs) and similar devices. Firstly, a short overview of the written code is provided in subsection 1.1. Then, the solver is compared to an analytical solution in subsection 2.1. Afterwards, the IV-characteristic of a RTD is calculated using the solver in subsection 2.2. Finally, some similar devices are studied in subsection 2.3.

1.1 Code Description

To solve the time-dependent Schrödinger equation in RTDs and similar devices, a Python class was developed in the Python file **Class.RTD.py**. Comparison with an analytical solution, calculation of the IV-characteristic and the experiments were performed in the Python files **RTD_analytical.py**, **RTD_IV.py** and **RTD_3barriers.py** which makes use of the developed Python class. A short overview of the most important functions of this class are given in table 1.

Name	Arguments	Short description
add_barriers	(U0)	Adds two potential barriers with height U0 at positions defined in project assignment
add_potential	(V0)	Puts a potential V0 over the device, as defined in project assignment
plot_potential	None	Plots the real and imaginary parts of the energy potential
add_recorder	(xr)	Puts a recorder at position x=xr
show_recorder	None	Plots the recorded values of ψ
update_loop	None	Solves problem for all timesteps Nt
animate	None	animates the probability density over time
analytical.T	(E.max)	Returns analytical transmission coefficient T with its corresponding energies E, with E going from 0 to E.max
J.time	None	Calculates the probability current density of a performed simulation in the time domain
J.freq	None	Calculates the probability current density of a performed simulation in the energy domain
restart	None	Sets all fields to zero

Table 1: Most important functions of class RTD

2 Results and Discussion

2.1 Validation

To verify our solver, a comparison with an analytical solution is made. The analytical solution provided in the project assignment uses a plane wave model to calculate the transmission coefficient through the RTD structure for a single energy. By using the transfer matrix method, an analytical expression for the transmission coefficient at zero applied voltage $T(E)$ can be found, where $T(E)$ is defined as the ratio of quantum probability densities:

$$T(E) = \frac{j_{x,trans}(E)}{j_{x,inc}(E)} \quad (1)$$

Where $j_{x,trans}$ is the quantum current that passed through the 2 barriers, and $j_{x,inc}$ the current when no barriers are present. This can be calculated numerically by the solver by using the functions described in subsection 1.1, both in the second order and fourth order schemes. The numeric calculations, performed in **RTD_analytical.py**, are then compared to the analytical solution in figure 1.

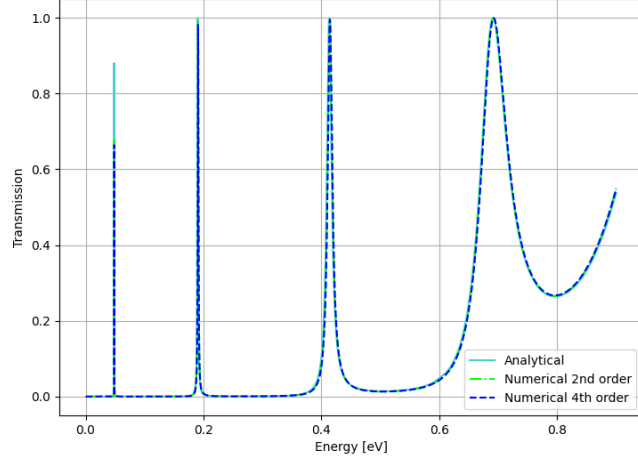
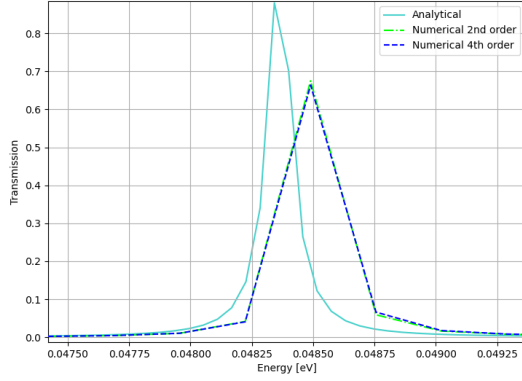
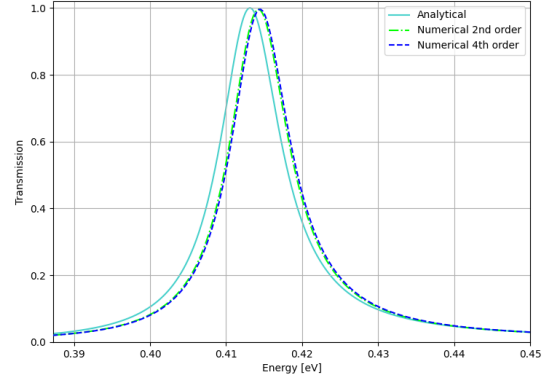


Figure 1: Comparison of analytical and numeric solutions

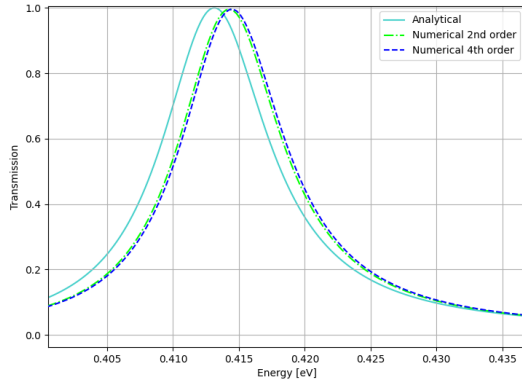
It can clearly be seen that both schemes reproduce the analytical solution well. For a zoomed in image of the peaks, figure 2 can be consulted. In this figure, the peaks of figure 1 are shown from left to right.



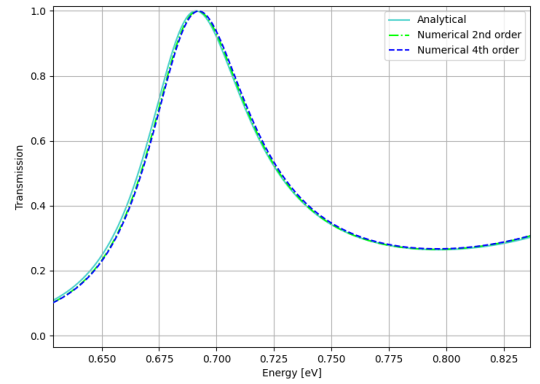
(a) First peak



(b) Second peak



(c) Third peak



(d) Fourth peak

Figure 2: Zoomed in images of different peaks of the transmission coefficient for 2 potential barriers

It can be seen that the first peak is not reproduced perfectly. This is because this peak is very narrow and needs a lot of sample points to be reproduced. This would need a lot of time samples, which would increase the computation time drastically. If the reader would like to resolve this peak, they could run the code **RTD_analytical.py** on e.g. a supercomputer with a very large t_{max} .

To study the difference between the second and fourth order scheme more clearly, a similar simulation as above has been run with parameters $N_x = \frac{L_x}{\Delta x} = 300$ and $t_{max} = 80L_x\sqrt{\frac{m^*}{2E}}$ (and some different parameters for the ABCs to ensure optimal working conditions). The resulting third and fourth peak are shown in figure 4. Here, it can be seen that the fourth order scheme consistently lays closer to the analytical peak position than the second order scheme, thus better reproducing the analytical result. The first and second peak are not shown since they need more time steps for better resolution.

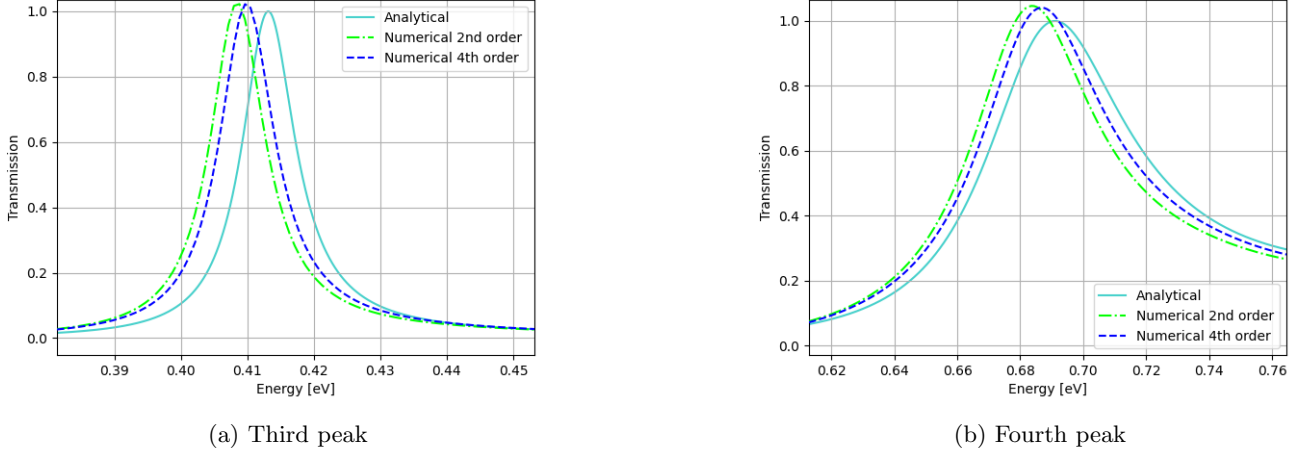


Figure 3: Zoomed peaks for comparison second and fourth order

2.2 Calculation of the IV-characteristic

Once our solver is validated, it can be used to calculate the IV-characteristic of an RTD. This is done in the python file **RTD_IV.py**. From Landauer theory, it is known that the current can be calculated as follows:

$$I(V_{DC}) = \frac{2e}{h} \int_{-\infty}^{+\infty} \sum_{nm} T_{nm}(E, V_{DC}) [f_L(E) - f_R(E)] dE \quad (2)$$

Here, $T_{nm}(E, V_{DC})$ is the transmission function as calculated in section 2.1, and $f_L(E)$ and $f_R(E)$ are the well-known Fermi-Dirac distributions for the left and right lead, respectively. In addition, it is known from theory that changing nm will result in a shift in the transmission coefficient: $T_{nm}(E) = T_{00}(E - E_{nm})$. Exploiting this knowledge, the transmission coefficient should only be calculated for one nm combination at each voltage V_{DC} , saving a lot of computational time.

In figure 4a, the IV curve is shown at different temperatures. As spoiled in the assignment, a region with negative differential resistance can be observed: around 50 mV, the current through the RTD starts to drop. Around 80 mV, the same effect is present. At high temperatures, the negative resistance effect is more pronounced as the current drops faster. On the other hand, a higher temperature comes with higher leakage currents.

In figure 4b, the zone of interest has been restricted to the first peak at room temperature, with a high resolution. It can be seen that the current is quite constant, and then starts to increase rapidly like a diode. However, the current reaches a peak and then decreases, resulting in a negative resistance. This property can be used e.g. for oscillators, amplifiers or switching circuits.

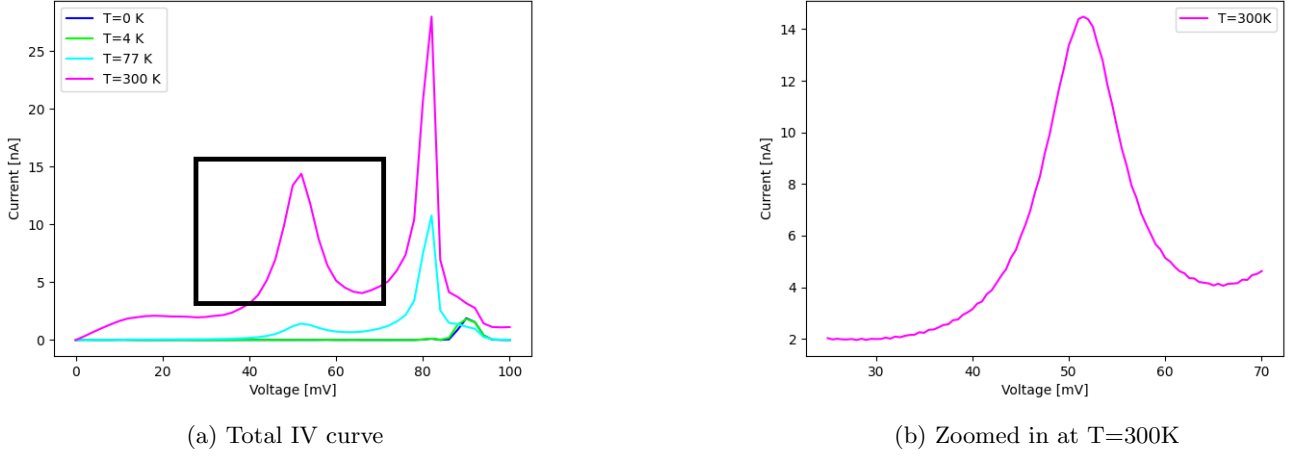


Figure 4: IV curve at different temperatures

2.3 Effects of extra barriers

In this section, the result of a third barrier will be discussed. A third, new barrier, with width $b = 5$ nm, is placed a distance $a = 15$ nm from the back of the second one. All calculations in this section were performed in the python file **RTD_3barriers.py**.

The transmission coefficient can again be calculated numerically and analytically for both 2nd and 4th order schemes. The result of these calculations can be found in figure 5, with zoomed in versions in figure 6. It is clear that adding a third barrier results in peak doubling due to coupling between the two quantum wells. Each original resonant state can now exist in two coupled configurations (a doublet): a symmetric state and an antisymmetric state. This phenomenon is analogous to the splitting of atomic orbitals in a diatomic molecule. Based on this logic, a fourth barrier would give rise to triplet states. This effect is shown in figure 7. In general, N barriers would lead to $N-1$ quantum wells and thus $N-1$ states per original resonant state.

It is again very hard to resolve the first peaks due to their extremely narrow nature. To this end, an IV-characteristic is not calculated for the 3 barrier case, as it would take too long. However, code that could construct an IV-characteristic via, *e.g.* a supercomputer is given in the **RTD_3barriers.py** file.

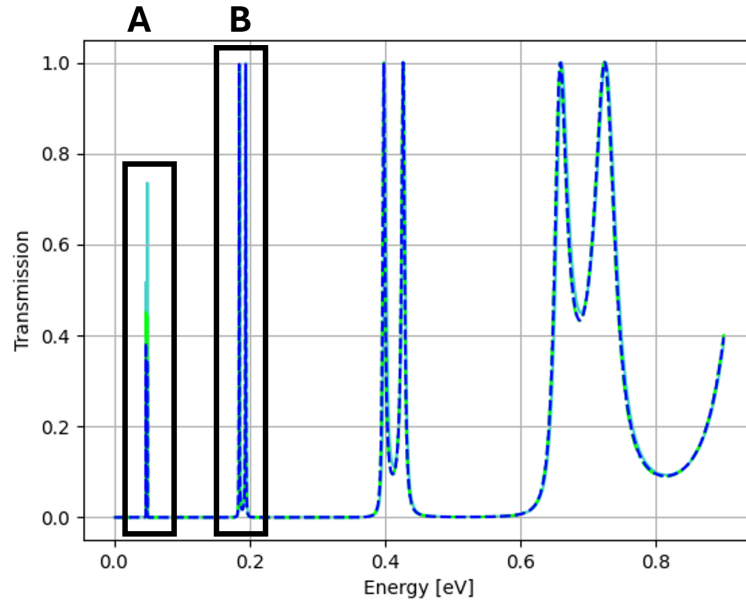
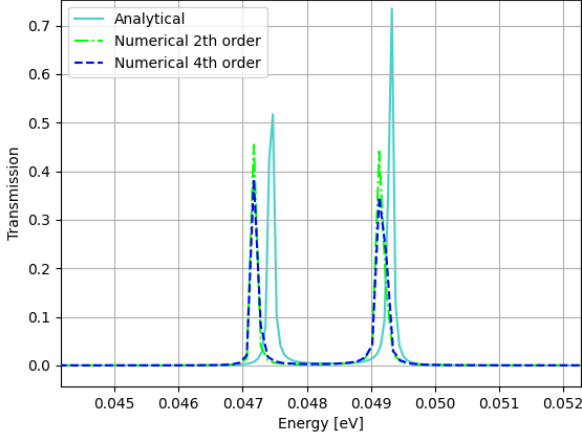
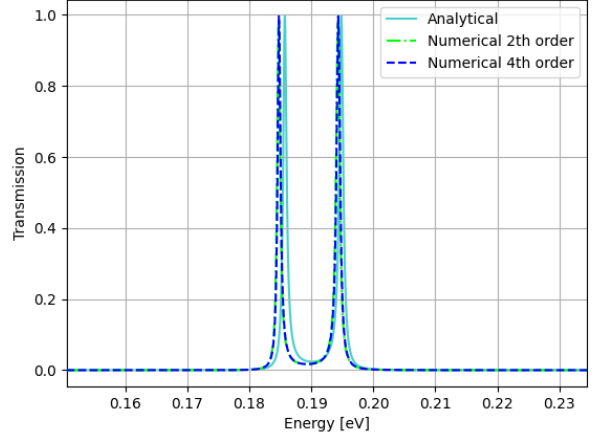


Figure 5: Transmission coefficient for 3 potential barriers

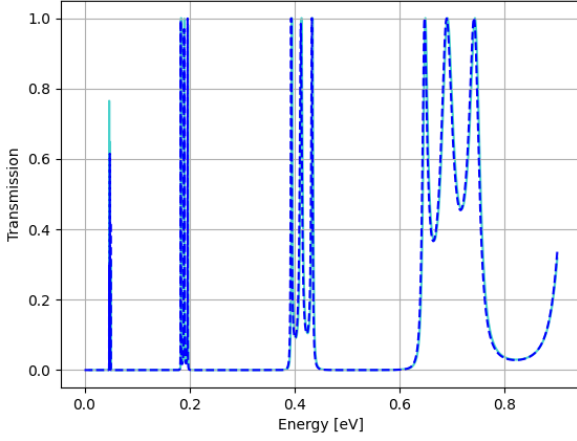


(a) First peaks A

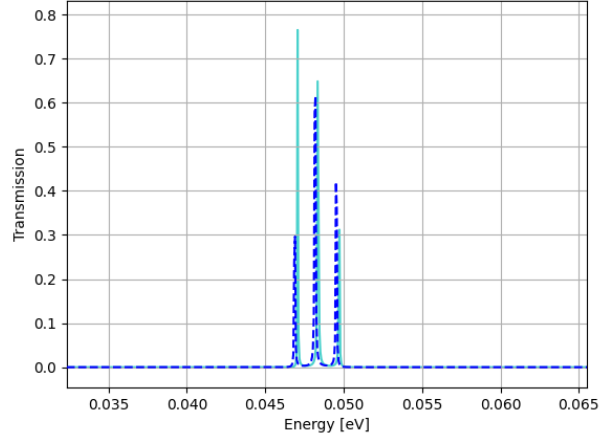


(b) Second peaks B

Figure 6: Zoomed in images of different peaks of the transmission coefficient for 3 potential barriers



(a) Total E domain



(b) Zoomed in on first triplet

Figure 7: Transmission coefficient for 4 potential barriers

3 Conclusion

In this project, a python code was successfully created to simulate a resonant tunneling diode and similar devices, and to extract numerical results such as the transmission coefficient and the IV-characteristic.

The transmission coefficient corresponds well to the analytical solution, both for the 2nd and 4th order schemes, although the peaks at low energies are not fully resolved. This can be overcome by simulating more spatial and temporal steps, but this has a high computational cost.

The IV curve shows a region with negative differential resistance, as expected. At high temperatures, this negative resistance has a larger effect, but this comes with a higher leakage current.

Other devices with three or more barriers were also simulated. The coupling of the different quantum wells gives rise to the splitting of the transmission coefficient peaks. Due to the long computation time an IV curve was not extracted for these devices.